

# PIT KS — SDSS

Parametric

Nonparametric

Foundation

|      | Parametric         |                    |                    |                 |                  |                    |                     |                       |                    |                      |                        |                 | Nonparametric      |                    |                 |                    |                    |                    |                    |                    | Foundation         |                    |                    |  |
|------|--------------------|--------------------|--------------------|-----------------|------------------|--------------------|---------------------|-----------------------|--------------------|----------------------|------------------------|-----------------|--------------------|--------------------|-----------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--|
| n    | LinearGauss-Homo   | LinearGauss-Hetero | Student-t          | LogNormal-Homo  | LogNormal-Hetero | Gamma-GLM          | LinGauss-Homo-Ridge | LinGauss-Hetero-Ridge | Student-t-Ridge    | LogNormal-Homo-Ridge | LogNormal-Hetero-Ridge | Gamma-GLM-Ridge | FlexCode-RF        | BART-Homo          | BART-Hetero     | Flow-Spline        | Quantile-Tree      | CatMLP             | MDN                | TabPFN Native      | TabPFN 2.5         | RealTabPFN 2.5     | TabICL Quantiles   |  |
| 500  | 0.1<br>(8.0e-3)    | 0.2<br>(9.2e-3)    | 0.1<br>(8.9e-3)    | 0.1<br>(1.2e-2) | 0.2<br>(7.4e-3)  | 0.1<br>(1.1e-2)    | 0.1<br>(8.0e-3)     | 0.2<br>(9.7e-3)       | 0.1<br>(1.2e-2)    | 0.1<br>(8.4e-3)      | 0.2<br>(6.4e-3)        | 0.2<br>(2.7e-2) | 0.1<br>(1.4e-2)    | 0.2<br>(1.4e-2)    | 0.2<br>(1.1e-2) | 9.4e-2<br>(1.1e-2) | 0.1<br>(4.3e-3)    | 0.1<br>(1.4e-2)    | 0.1<br>(1.2e-2)    | 9.4e-2<br>(6.2e-3) | 9.4e-2<br>(6.2e-3) | 8.8e-2<br>(6.8e-3) | 7.9e-2<br>(5.0e-3) |  |
| 1K   | 0.1<br>(2.0e-2)    | 0.2<br>(2.5e-2)    | 8.0e-2<br>(2.1e-2) | 0.1<br>(1.9e-2) | 0.2<br>(1.2e-2)  | 0.1<br>(1.9e-2)    | 0.1<br>(2.1e-2)     | 0.2<br>(2.5e-2)       | 8.2e-2<br>(2.0e-2) | 0.1<br>(1.8e-2)      | 0.2<br>(9.6e-3)        | 0.1<br>(1.8e-2) | 9.0e-2<br>(5.2e-3) | 0.2<br>(1.3e-2)    | 0.1<br>(2.8e-3) | 4.9e-2<br>(8.0e-3) | 6.8e-2<br>(2.1e-2) | 5.7e-2<br>(2.4e-5) | 6.9e-2<br>(1.5e-2) | 5.3e-2<br>(5.6e-3) | 5.3e-2<br>(5.6e-3) | 4.5e-2<br>(4.0e-3) | 4.9e-2<br>(8.0e-3) |  |
| 10K  | 9.4e-2<br>(3.0e-3) | 0.2<br>(1.6e-4)    | 3.2e-2<br>(9.0e-3) | 0.1<br>(1.0e-2) | 0.2<br>(1.7e-3)  | 8.5e-2<br>(8.1e-3) | 9.4e-2<br>(3.0e-3)  | 0.2<br>(3.0e-4)       | 3.1e-2<br>(9.0e-3) | 0.1<br>(1.0e-2)      | 0.2<br>(1.7e-3)        | 0.1<br>(1.1e-2) | 1.0e-1<br>(2.7e-3) | 9.0e-2<br>(1.0e-2) | 0.2<br>(8.8e-3) | 4.3e-2<br>(2.9e-2) | 2.4e-2<br>(4.5e-3) | 6.7e-2<br>(6.9e-3) | 2.8e-2<br>(1.7e-3) | 2.0e-2<br>(3.9e-3) | 2.0e-2<br>(3.9e-3) | 2.0e-2<br>(7.6e-4) | 1.8e-2<br>(3.1e-3) |  |
| 50K  | 9.5e-2<br>(3.7e-3) | 0.1<br>(3.6e-3)    | 2.8e-2<br>(1.9e-3) | 0.1<br>(1.0e-3) | 0.2<br>(1.6e-4)  | 9.3e-2<br>(9.7e-4) | 9.5e-2<br>(3.6e-3)  | 0.1<br>(3.7e-3)       | 2.9e-2<br>(1.8e-3) | 0.1<br>(1.0e-3)      | 0.2<br>(3.1e-4)        | 0.1<br>(3.5e-3) | 0.1<br>(1.4e-4)    | 8.7e-2<br>(6.0e-4) | 0.1<br>(3.8e-3) | 2.0e-2<br>(9.5e-3) | 1.9e-2<br>(3.9e-3) | 3.8e-2<br>(5.3e-3) | 1.9e-2<br>(4.0e-3) | 2.4e-2<br>(3.2e-3) | 2.4e-2<br>(3.2e-3) | 1.6e-2<br>(3.2e-3) | —                  |  |
| 100K | 9.4e-2<br>(2.8e-3) | 0.1<br>(1.7e-3)    | 2.9e-2<br>(5.6e-4) | 0.1<br>(1.5e-3) | 0.2<br>(4.0e-3)  | 8.8e-2<br>(1.5e-3) | 9.4e-2<br>(2.7e-3)  | 0.1<br>(1.7e-3)       | 2.9e-2<br>(5.8e-4) | 0.1<br>(1.5e-3)      | 0.2<br>(4.0e-3)        | 0.1<br>(7.2e-4) | 0.1<br>(1.7e-3)    | 8.6e-2<br>(4.0e-4) | 0.1<br>(1.9e-3) | 2.1e-2<br>(7.1e-3) | 1.7e-2<br>(1.4e-3) | 2.8e-2<br>(4.4e-3) | 1.5e-2<br>(2.7e-3) | —                  | —                  | —                  | —                  |  |
| 250K | 9.2e-2<br>(1.4e-4) | 0.1<br>(6.6e-4)    | 2.8e-2<br>(5.8e-5) | 0.1<br>(4.4e-5) | 0.2<br>(4.6e-4)  | 8.6e-2<br>(5.9e-5) | 9.2e-2<br>(1.3e-4)  | 0.1<br>(6.8e-4)       | 2.8e-2<br>(3.9e-5) | 0.1<br>(5.0e-5)      | 0.2<br>(4.4e-4)        | 0.1<br>(2.1e-4) | 0.1<br>(1.0e-3)    | 8.7e-2<br>(7.3e-5) | —               | 3.5e-2<br>(2.1e-2) | —                  | 2.7e-2<br>(1.7e-3) | 1.8e-2<br>(2.8e-3) | —                  | —                  | —                  | —                  |  |
| 500K | 9.3e-2<br>(3.6e-4) | 0.1<br>(8.2e-4)    | 2.9e-2<br>(8.4e-4) | 0.1<br>(9.1e-4) | 0.2<br>(1.8e-4)  | 8.7e-2             | 9.3e-2              | 0.1                   | 3.0e-2             | 0.1                  | 0.2                    | 0.1             | 0.1                | 9.1e-2<br>(6.3e-4) | —               | 3.3e-2<br>(2.1e-2) | —                  | 2.8e-2             | 2.1e-2             | —                  | —                  | —                  | —                  |  |